Secant Method

Step 1: Input Parameters

- Choose two initial guesses: x0 and x1.
- Set the maximum number of iterations: step.
- Set the error tolerance: epes.

Step 2: Check for a Root in the Interval

- If f(x0) ×f(x1) ≥ 0, then there is no guarantee of a root in the interval. Display a warning and stop execution.

Step 3: Initialize Variables

- Set iteration counter n=1n = 1n=1.
- Initialize error er=1.

Step 4: Iterate Until Convergence

- While n ≤ step and er> epes:
    1. Compute f(x0) and f(x1).
    2. Check if f(x1) −f(x0) =0. If true, stop execution to prevent division by zero.
    3. Compute the next approximation using the **Secant Formula**:

$$x2=x1-\frac{f(x1)*(x1-x0)}{f(x1)-f(x0)}$$

    4. Calculate relative error: er=|x2−x1/x2|
    5. Update values:
        - Set x0=x1
        - Set x1=x2
        - Increment n.

Step 5: Output the Result

- If the loop exits, return x2 as the approximate root.
- If the maximum iterations are reached without convergence, display an error message.

Example of the Algorithm in Action

Given:

f(x)=x2−4, with initial guesses x0=1 and x1=3

Iteration Process:

1. Compute f (1) =−3 and f (3) =5 Since f (1) ×f (3) <0, a root exists.
2. Use the secant formula to compute x2

$$x2=x1-\frac{f(x1)*(x1-x0)}{f(x1)-f(x0)}$$

$$x2=3-\frac{5*(3-1)}{5-(-3)}$$

$$x2=3-\frac{10}{8}$$

$$x2=3-1.25$$

$$x2=1.75$$

now,

x0=x1 x0=3

x1=x2 x1=1.75

again, using secant formula, then find next x2

3. Check error and update values.
4. Repeat until convergence.